

A	2
9	X

FIGURE 6.1 Classic Confirmation Bias Experiment

Technical Description

Confirmation bias can be thought of as a form of selection bias in collecting evidence for supporting certain beliefs, whereby decision makers observe, overvalue, or actively seek out information that confirms their claims, while simultaneously ignoring or devaluing evidence that might discount their claims. A classic demonstration of confirmation bias, of which there are many versions, is one in which subjects are shown four cards, each with a number on one side and a letter on the other. They are then told the following rule: “If the card has a vowel on one side, then it *must* have an even number on the other side.” The cards are then laid out as depicted in Figure 6.1. Subjects are then asked, “Which two cards would you turn over to test the rule?”

When this experiment is run, most participants do not choose the correct cards (the card reading “A” and the card reading “9”). Instead, the most frequent responses are “A” and “2.” This pairing demonstrates a common logical fallacy: People choose “2” because the discovery of an accompanying vowel could indeed uphold the hypothesis. However, exposing the opposite side of the “2” card can’t possibly invalidate the hypothesized condition, so this can’t be the correct response. People more readily identify “2” rather than “9” because confirmation bias makes them *want* to validate the hypothesis—not refute it as directed.

Another lesson here is that beliefs don’t need to be logically entrenched in order to kindle confirmation bias. The hypothesis as stated becomes an immediate if subtle aspect of the participant’s choice of cards. Even though subjects have no reason to accept the hypothesis, they become loyal enough to its validity that they are unable to recognize the correct answer choice. In fact, when beliefs *are* firmly established in evidence, the effects of confirmation bias become less overt. This is due to a tendency to give more attention and weight to data that fit with beliefs with stronger foundations.

Numerous studies have demonstrated that people excessively value confirmatory information, that is, positive or supportive data. The “most likely